

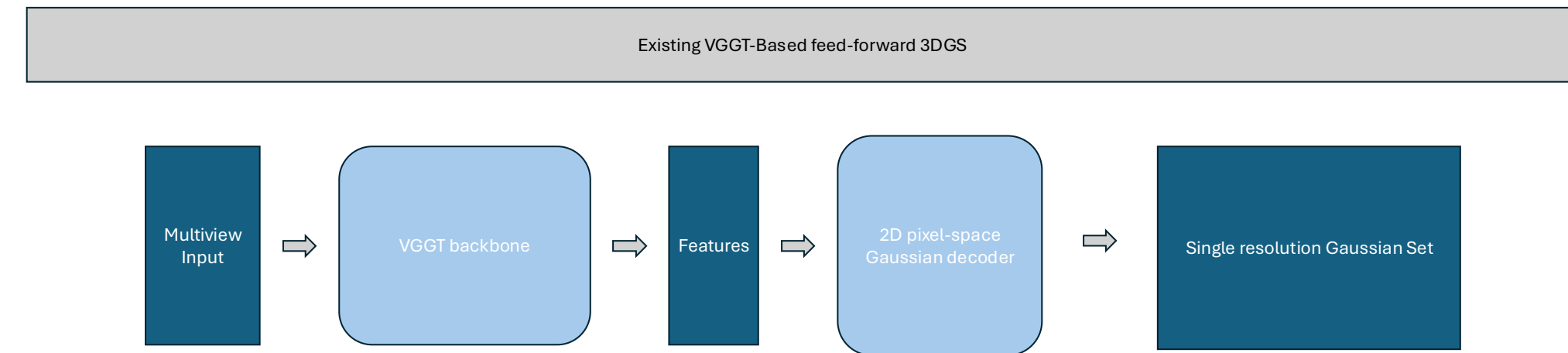
Motivation

VGGT: Already provides 3D-aware features, depth, and camera poses from the input views.

Existing VGGT-based methods: decode Gaussians in 2D image space, one gaussian per pixel.

This leaves a mismatch:

- 3D structure is inferred by the backbone, but not used by the decoder.
- It also limits multi-resolution generation, since a flat 2D decoder does not naturally produce coarse-to-fine Gaussian layers.



Key Idea

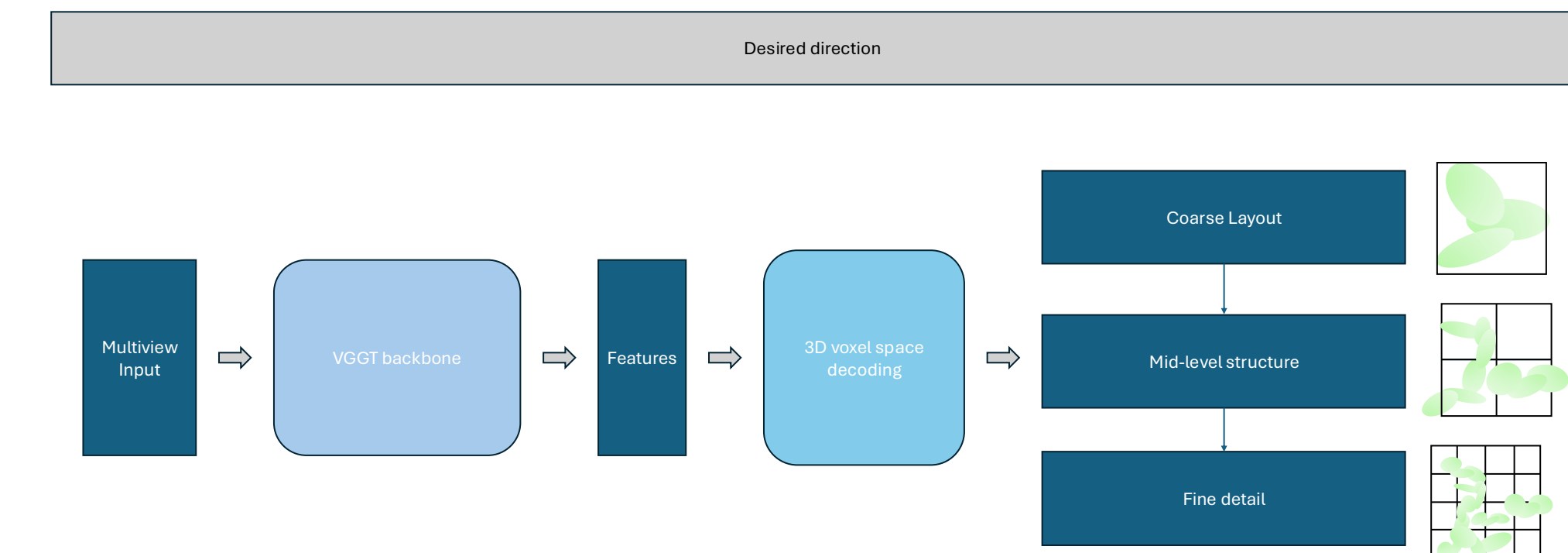
SplatTree lifts VGGT features into a multi-resolution 3D voxel grid, then decodes Gaussians with sparse 3D convolutions.

The decoder produces a multi-resolution Gaussian tree:

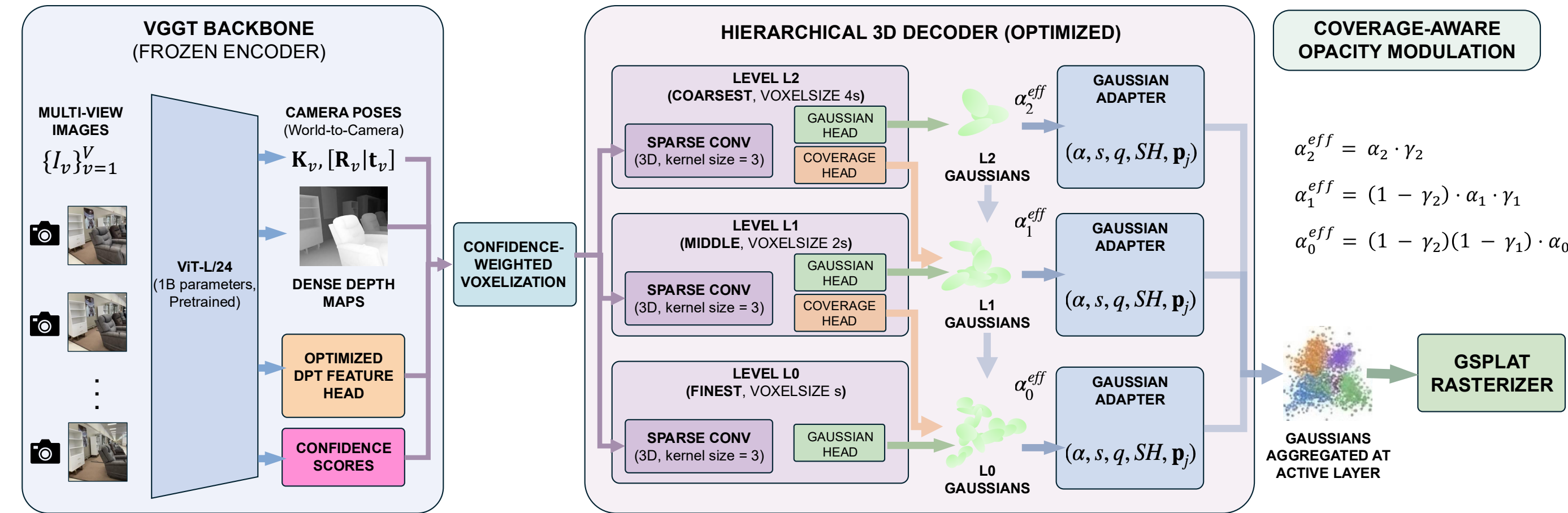
- L2: coarse scene layout
- L1: object structure and boundaries
- L0: fine texture and local detail

Coverage heads decide which layer should explain each region, so coarse Gaussians handle simple areas and finer Gaussians refine complex ones.

Decode once in 3D. Render at the level you need.



Approach



1. Build 3D voxel features

Sparse input views are processed by VGGT and lifted into a shared 3D voxel feature grid.

2. Decode at three scales

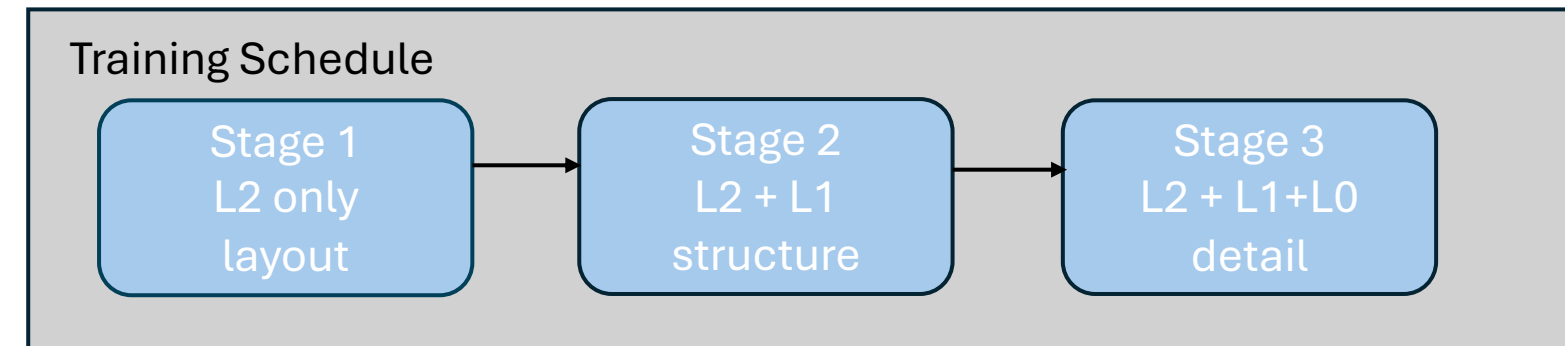
The decoder predicts Gaussians at three voxel resolutions: L2 for coarse layout, L1 for middle-level structure, and L0 for fine detail.

3. Predict Gaussians

Sparse 3D convolutions predict parameters from local 3D voxel neighborhoods.

4. Blend layer

Coverage values modulate Gaussian opacity so coarse layers explain simple regions and finer layers refine the remaining details.



Layer effect

Each layer adds a different level of detail.

L2 only captures a coarse scene layout with a small Gaussian count.

+L1 recovers object structure and sharper boundaries.

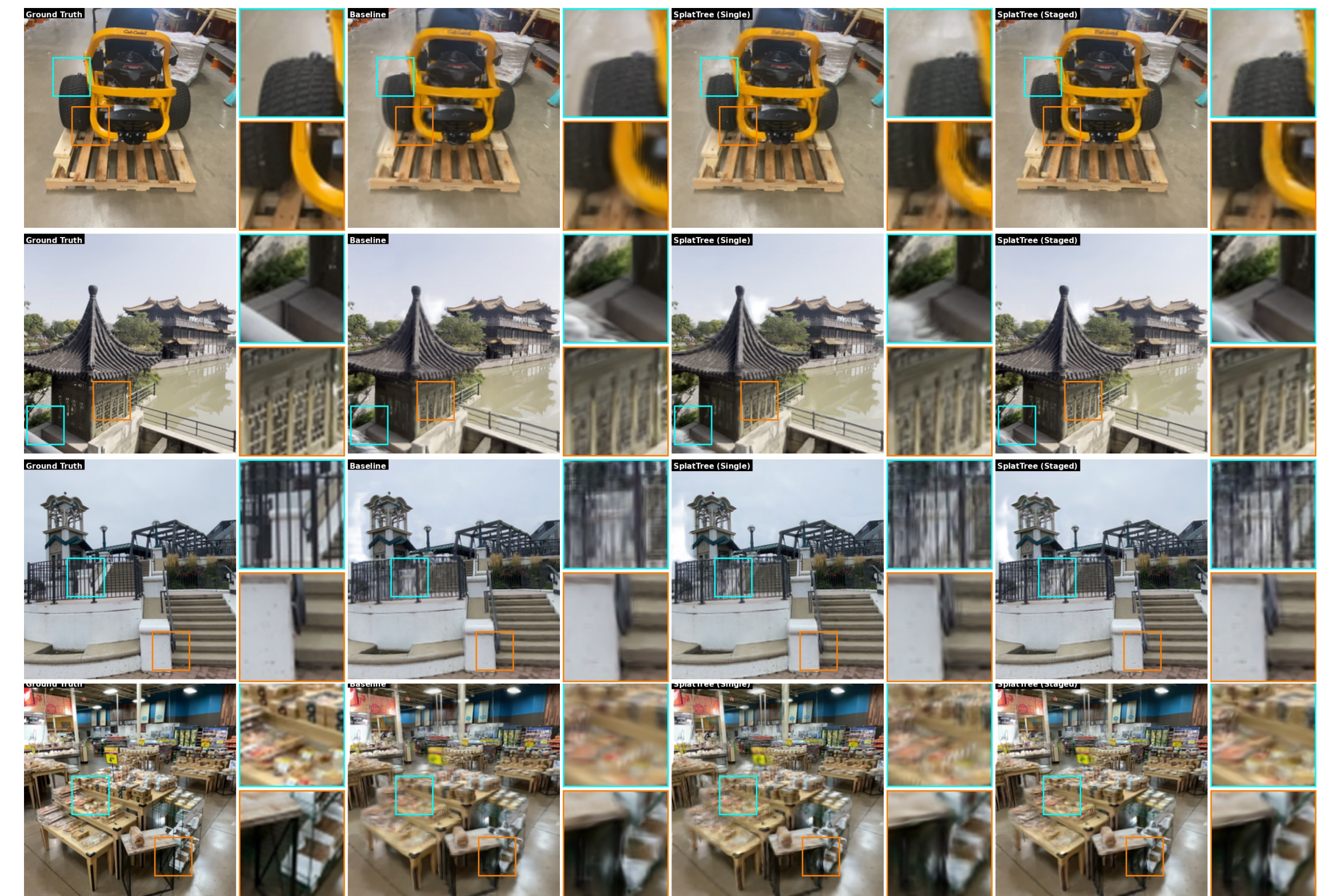
+L0 further improves fine texture and local details.

Layers Rendered	PSNR↑	SSIM↑	LPIPS↓	#GS
L2 only (coarsest)	15.83	0.512	0.610	61K
L1 + L2	21.04	0.686	0.351	225K
L0 + L1 + L2 (all)	23.65	0.762	0.233	574K
<i>Improvement per layer:</i>				
L2 → L1+L2	+5.21	+0.174	-0.259	+164K
L1+L2 → All	+2.61	+0.076	-0.118	+349K

Results

SplatTree improves rendering quality over 2D pixel-wise decoding under the same VGGT backbone, while its multi-resolution layers provide a clear quality and cost tradeoff.

Method	Steps	PSNR↑	SSIM↑	LPIPS↓	#GS
AnySplat	20K	23.01	0.743	0.258	352K
SplatTree-Single	20K	23.47	0.758	0.242	350K
SplatTree	40K	23.65	0.762	0.233	574K



Conclusion

SplatTree decodes Gaussians in 3D voxel space instead of 2D image space. This improves rendering quality by using local 3D context.

Its multi-resolution Gaussian tree enables progressive rendering from coarse layout to fine detail.

Takeaway:

3D decoding makes feed-forward Gaussians both spatially aware and resolution adaptive.